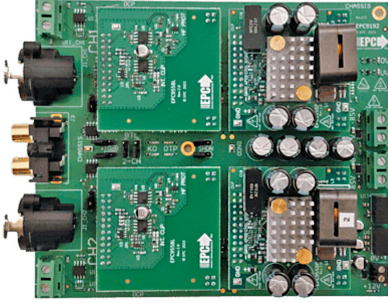
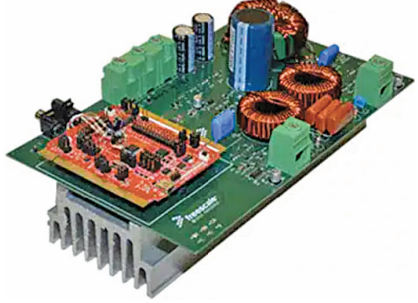


Interesting

Reference Designs OF AUDIO AMPLIFIERS

Title of Reference Design	Two-Channel Audio Amplifier	Class-D Audio Amplifier
Image		
Key Attractions	The EPC9192 board features a two-channel Class-D amplifier, delivering 700W per channel at 4Ω, and doubling to 1400W at 8Ω in BTL mode. It includes scalable, modular components.	The 1kW Class-D audio amplifier features a push-pull power converter, rapid prototyping capabilities, and PWM control through an internal timer. It includes protective measures to ensure operational safety and CPU efficiency.
Highlights	The EPC9192 evaluation board by Efficient Power Conversion (EPC) utilises a ground-referenced, split dual supply in a single-ended (SE) configuration with EPC's 200V, EPC2307 eGaN FETs in a half-bridge topology. It also supports a bridged-load (BTL) mode configuration, doubling the output to 1400W at 8Ω. The design is modular, featuring a mother-board with two PWM modulators and two half-bridge power stage daughterboards, making it scalable and adaptable for dual-channel amplification. It includes protective features and customisable PWM modulators to support various modulation techniques and switching frequencies. The power stages are designed for performance evaluation and comparative analysis. The kit contains two EPC9558L PWM modulator boards with an asynchronous, self-oscillating PWM modulator design, and EPC9558P power stages that integrate two EPC2307 eGaN FETs.	The 1kW Class-D audio amplifier reference design from NXP Semiconductors provides a comprehensive model for building an audio amplifier on the Kinetis KV1x Tower series platform, utilising a push-pull power converter. It uses the internal FlexTimer module to modulate analogue audio inputs into Class-D format and generate PWM signals to control the switching push-pull power supply. This design allows for rapid prototyping on either the tower system module or the freedom system platform, which is crucial for handling analogue audio input, producing Class-D audio output, and managing the push-pull power supply. The embedded source code facilitates the quick and cost-effective assembly of the Class-D audio amplifier, while the FlexTimer also manages the gate drivers for the power MOSFETs, integrating several protective features such as dead-time insertion, fault control, initialisation, and polarity control. These protections ensure operational safety and reliability, minimising CPU load to free up resources for further application enhancements.
Applications	The reference design suits applications requiring durable performance without the bulk and overheating.	The reference design suits applications where heat dissipation is a concern.
OEM Brand	Efficient Power Conversion (EPC)	NXP Semiconductors
URL	https://www.electronicsforu.com/electronics-projects/reference-design-for-2-x-700-w-4-%cf%89-class-d-amplifier	https://www.electronicsforu.com/electronics-projects/class-d-audio-amplifier-reference-design